SIMATS ENGINEERING

ASSESSMENT TOOL 1 – High (Coding Challenge)

COURSE CODE: CSA09 **COURSE NAME: Programming in Java**

COURSE OUTCOME COVERED

CO1: *Apply object-oriented programming principles such as classes, objects, encapsulation, data hiding, inheritance, and polymorphism to design and implement entity manipulations (BL3,BL4)*

QUESTIONS: 10 **Total Marks: 100**

ANNEXURE A **Coding Challenge**

Code of Conduct:

*I **B.ANJALI(192371062)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Coding Challenge

1. Library Book Management using Classes & Encapsulation

Design a Book class with private fields such as bookId, title, author, and price. Implement constructors, getters, setters, and methods to issue and return books. Create a menu-driven program to manage multiple books using an array of objects. Also, construct suitable test cases to validate book addition, issue, return, and search operations. BTL: BL3 (Apply)

2. Employee Payroll System using Inheritance

Design a payroll system with a base class Employee and derived classes PermanentEmployee and ContractEmployee. Override a method calculateSalary() in each subclass to compute salary differently. Display employee details and salary using runtime polymorphism. Also, construct suitable test cases to validate salary calculation for different employee types. BTL: BL3 (Apply)

3. Vehicle Rental System using Runtime Polymorphism

Develop a vehicle rental system with a superclass Vehicle and subclasses Car, Bike, and Bus. Override the method calculateRent(int days) in each subclass. Store vehicles in an array of Vehicle references and generate rental bills dynamically. Also, construct suitable test cases to validate rent calculation for different vehicle categories. BTL: BL4 (Analyze)

4. Bank Account System using Data Hiding

Design a BankAccount class with private fields accountNumber, holderName, and balance. Implement methods for deposit, withdrawal, and balance inquiry with proper validation. Prevent direct modification of balance from outside the class. Also, construct suitable test cases to validate valid and invalid transactions. BTL: BL3 (Apply)

5. University Admission System using Method Overloading

Create an Admission class that overloads the method calculateFee() for:

- Undergraduate students,
- Postgraduate students,
- Scholarship students.

Display the fee structure based on user input and compare the overloaded method executions. Also, construct suitable test cases to validate each overloaded version. BTL: BL3 (Apply)

6. Shape Area Calculator using Abstract Class & Polymorphism

Design an abstract class Shape with an abstract method area(). Implement subclasses Circle, Rectangle, and Triangle. Use an array of Shape references to compute and display areas dynamically. Also, construct suitable test cases to validate area calculations and polymorphic behavior. BTL: BL4 (Analyze)

7. Hospital Management System using Interfaces

Develop a hospital management system with an interface MedicalRecord containing methods addRecord() and displayRecord(). Implement the interface in classes Patient and Doctor.

Demonstrate interface-based abstraction and object interaction. Also, construct suitable test cases to validate record creation and display. BTL: BL3 (Apply)

8. Online Shopping Order System using Packages

Create two packages:

- shopping.product
- shopping.order

Implement classes Product and Order in separate packages. Import the packages into a main application that places orders and calculates the total amount. Also, construct suitable test cases to validate package usage, order placement, and bill generation. BTL: BL3 (Apply)

9. Student Result Analyzer using Arrays of Objects

Design a Student class containing roll number, name, and marks in three subjects. Store multiple student objects in an array and implement methods to calculate total, average, grade, and class topper. Also, construct suitable test cases to validate grading and topper identification. BTL: BL3 (Apply)

10. Smart Home Device Controller using Hierarchical Inheritance

Develop a smart home system with a base class Device and derived classes Light, Fan, and AirConditioner. Implement methods to turn devices on/off and display power consumption. Use polymorphism to control devices through a common reference. Also, construct suitable test cases to validate device operations and power calculations. BTL: BL4 (Analyze)

STUDENT NAME:B.Anjali

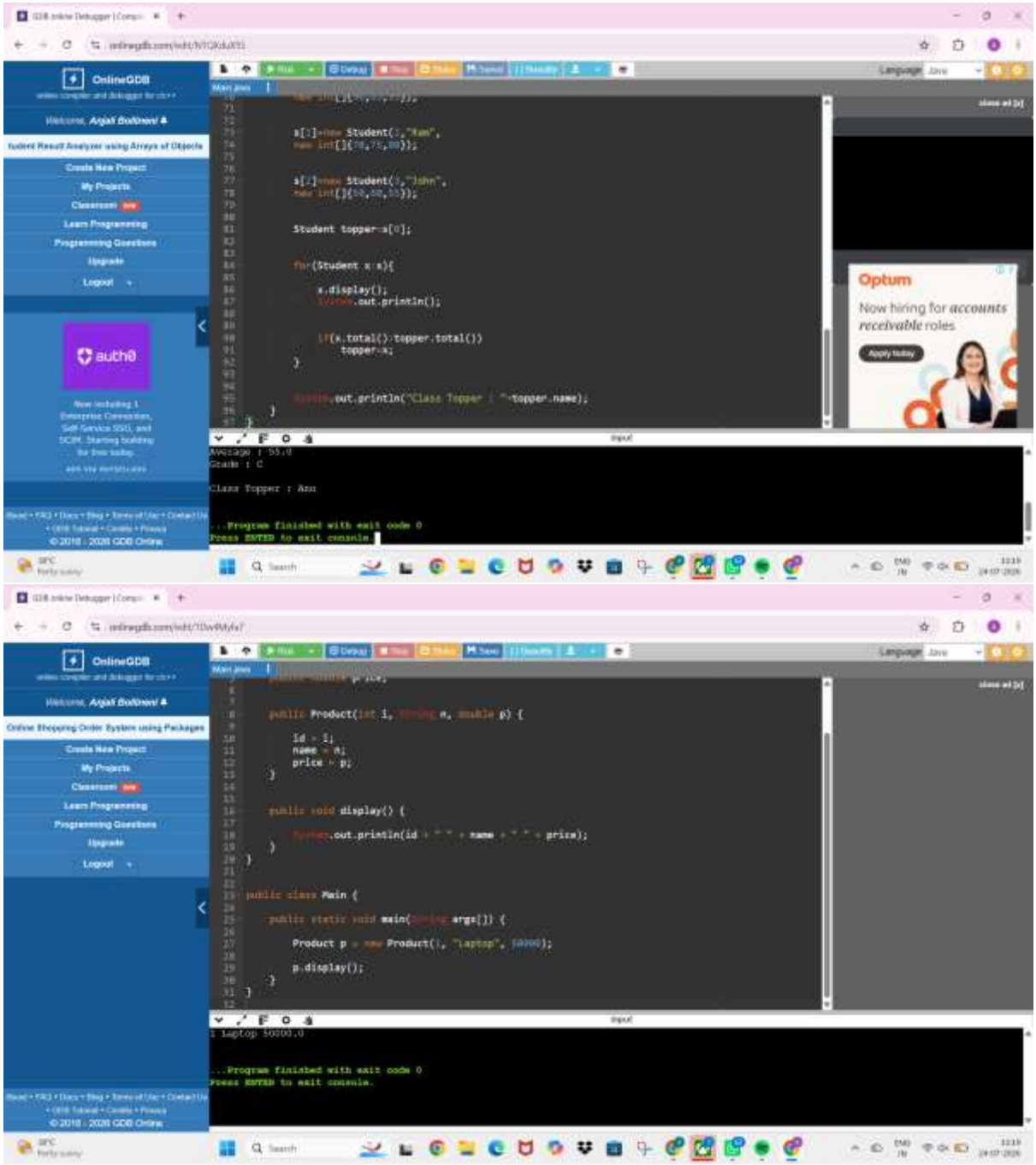
REGISTER NUMBER:192371062

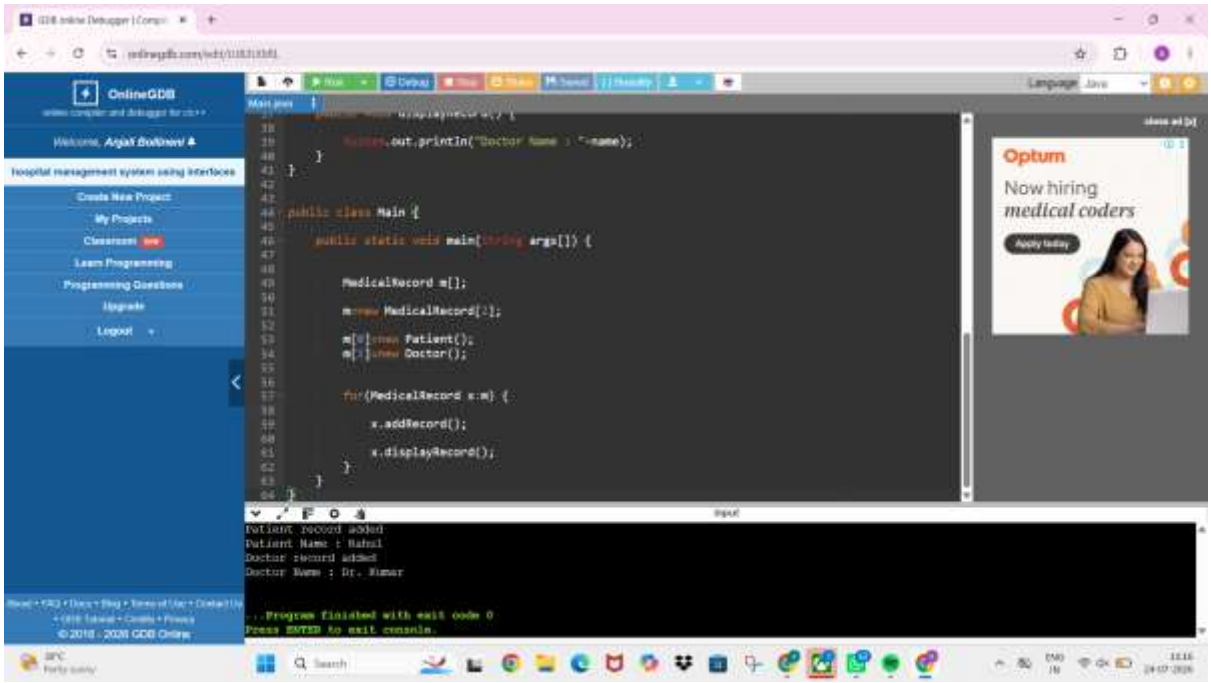
COURSE CODE: CSA0903

COURSE NAME: Programming in java

GUIDED BY: Magesh Kumar

DATE:24-07-2026

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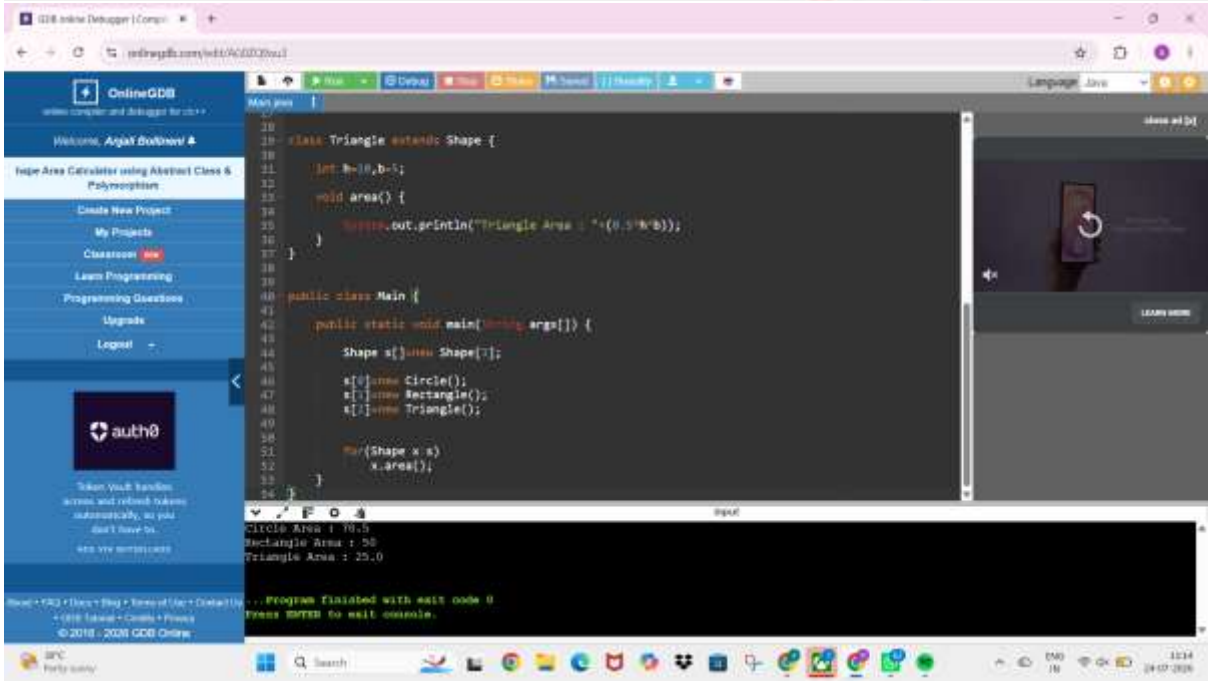
Optum
Now hiring
medical coders
Apply today
Patient record added
Patient Name : Rahul
Doctor record added
Doctor Name : Dr. Kumar
...Program finished with exit code 0
Press ENTER to exit console.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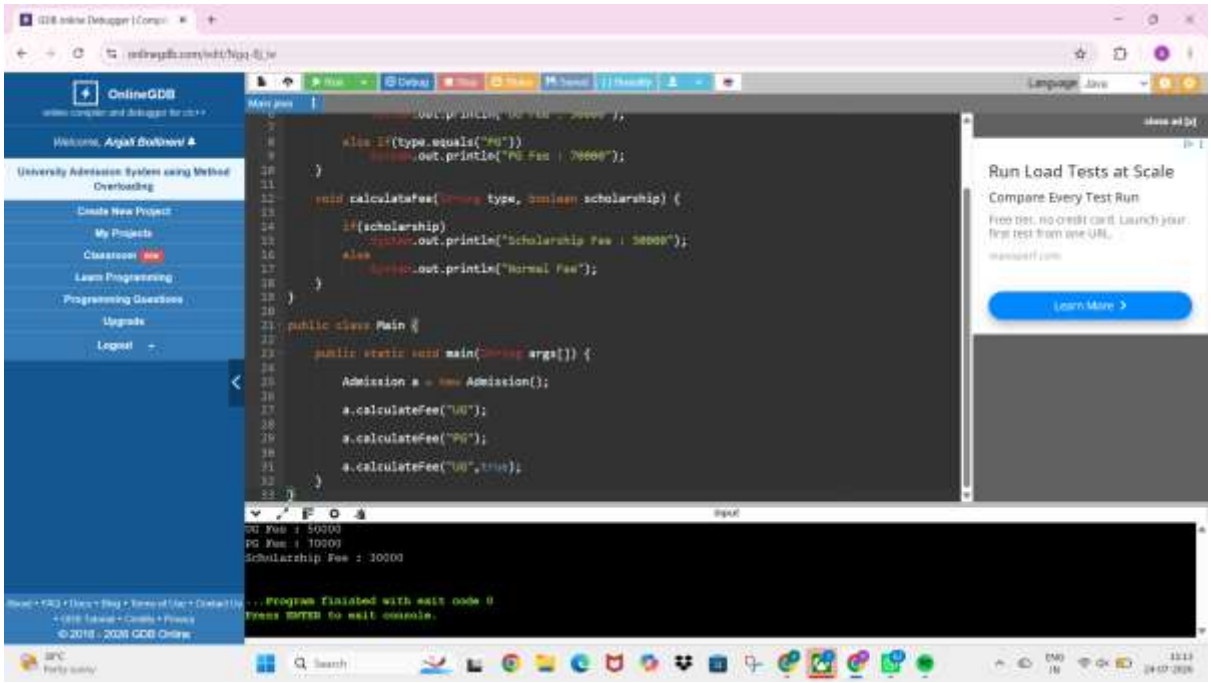
auth0
Main.java
Circle Area : 78.5
Rectangle Area : 50
Triangle Area : 25.0
...Program finished with exit code 0
Press ENTER to exit console.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Stop
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Save
Beautify
Language
Java
Input
UG Fee : 50000
PG Fee : 70000
Scholarship Fee : 30000
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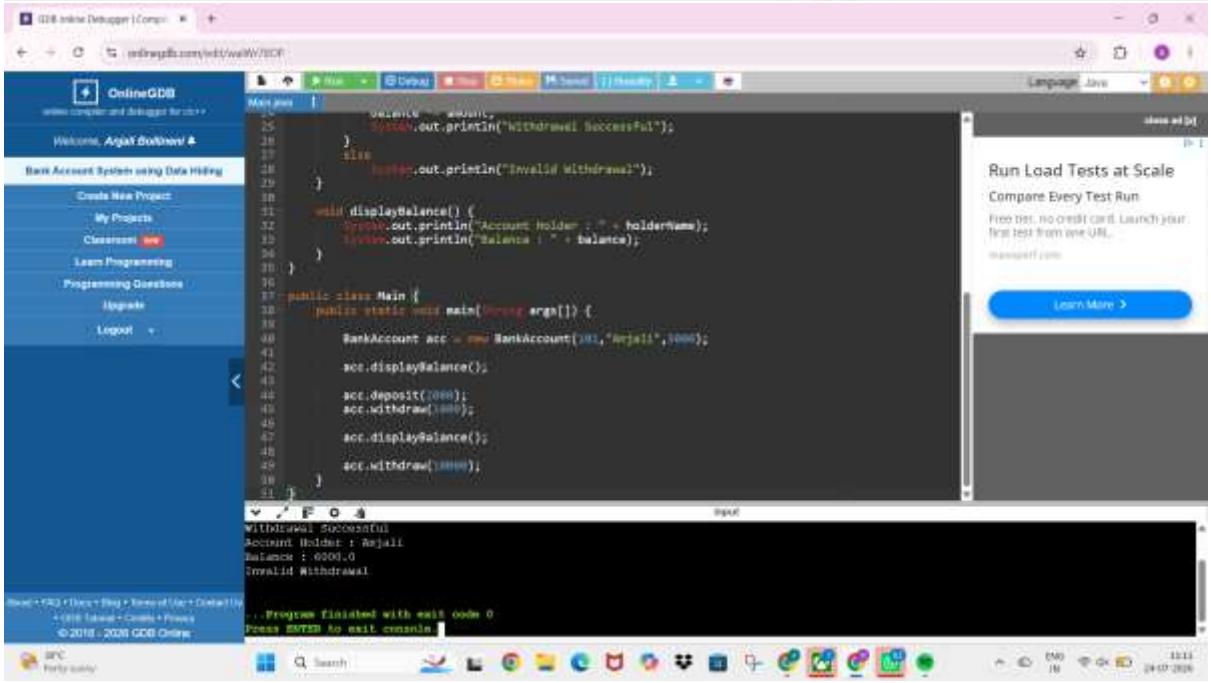
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Stop
Share
Save
Beautify
Language
Java
Input
Withdrawal Successful
Account Holder : Anjali
Balance : 6000.0
Invalid Withdrawal
...Program finished with exit code 0
Press ENTER to exit console.
Run Load Tests at Scale
Compare Every Test Run
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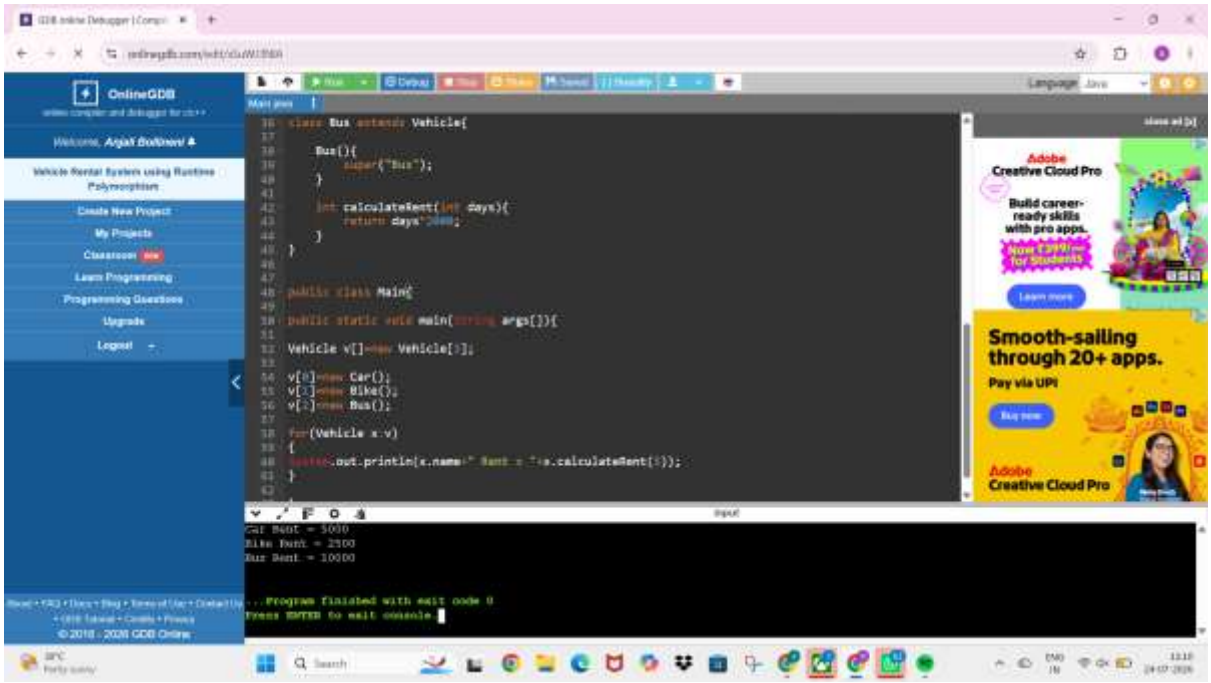
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Main.java
class Bus extends Vehicle{
Bus(){
super("Bus");
}
int calculateRent(int days){
return days*2000;
}
}
public class Main{
public static void main(String args[]){
Vehicle v[]=new Vehicle[3];
v[0]=new Car();
v[1]=new Bike();
v[2]=new Bus();
for(Vehicle x:v)
{
System.out.println(x.name+" Rent : "+x.calculateRent(5));
}
}
Car Rent : 5000
Bike Rent : 2500
Bus Rent : 10000
...Program finished with exit code 0
Press ENTER to exit console.
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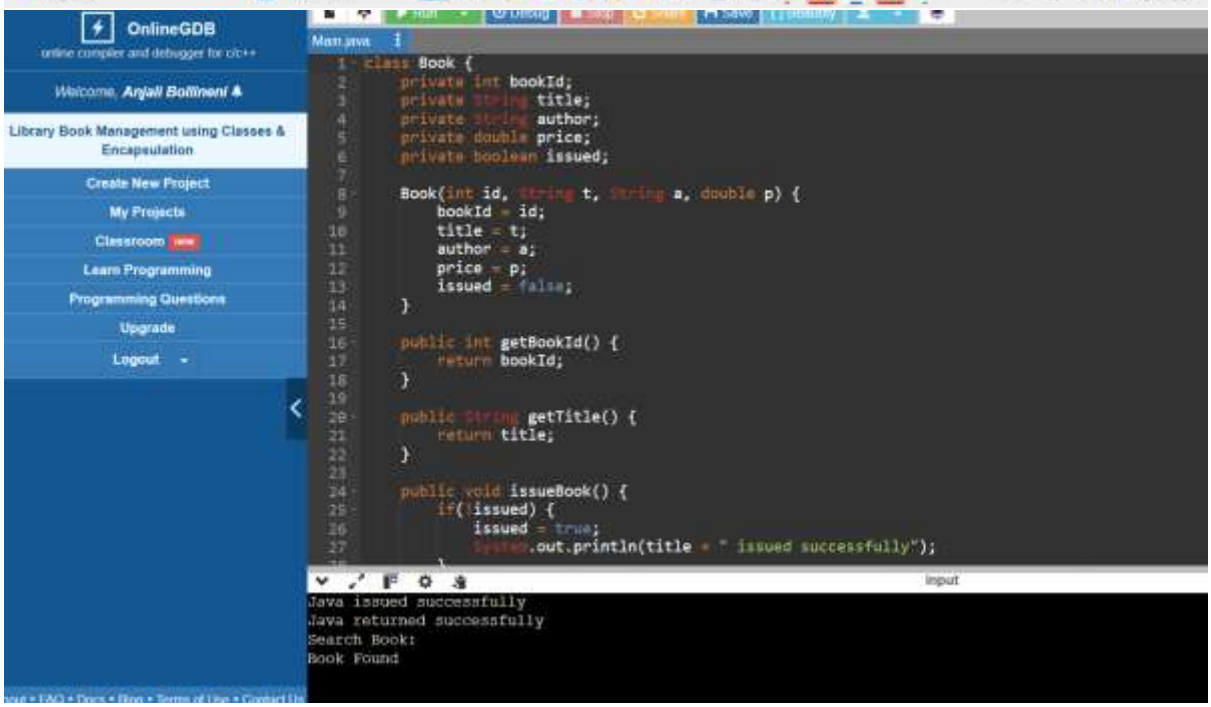
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Main.java
class Book {
private int bookId;
private String title;
private String author;
private double price;
private boolean issued;
Book(int id, String t, String a, double p) {
bookId = id;
title = t;
author = a;
price = p;
issued = false;
}
public int getBookId() {
return bookId;
}
public String getTitle() {
return title;
}
public void issueBook() {
if(!issued) {
issued = true;
System.out.println(title + " issued successfully");
Java issued successfully
Java returned successfully
Search Book:
Book Found

Main.java

```java
class ContractEmployee extends Employee{

    ContractEmployee(String n,int i){
        super(n,i);
    }

    double calculateSalary(){
        return 25000;
    }
}

public class Main{
    public static void main(String args[]){

        Employee e[];

        e=new Employee[2];

        e[0]=new PermanentEmployee("Anu",101);
        e[1]=new ContractEmployee("Ram",102);

        for(Employee x:e){
            x.display();
            System.out.println("Salary : "+x.calculateSalary());
        }
    }
}
```

input

```
Anu 101
Salary : 50000.0
Ram 102
Salary : 25000.0
```